

Test Progressive Assistant	
Business Analysis and Application	
For 25-2 Generation	
Periode Berlaku Semester Genap 2025/2026 Valid on Even Semester 2025/2026	Software Laboratory Center Academic Development

- Peserta TPA tidak diperkenankan untuk:
 - Memublikasikan jawaban dan/atau soal** sebelum waktu pelaksanaan TPA berakhir, baik dengan sengaja maupun tanpa disengaja.
 - Membuka dan/atau menyalin jawaban** dari buku, video, ataupun peserta lainnya.
 - Melakukan tindakan yang menyebabkan jawaban dicontek** oleh orang lain atau kelompok lain, baik disengaja maupun tidak disengaja.
 - Melakukan tindakan kecurangan lainnya.**
- Jika peserta TPA terbukti melakukan tindakan seperti yang dicantumkan pada butir ke-1, maka **nilai TPA** yang melakukan kecurangan, baik menyontek atau dicontek, akan **dinolkan** sesuai dengan peraturan yang berlaku. Dan akan diberikan **sanksi** sesuai prosedur yang berlaku.
- Jawaban yang dapat diterima dan dinilai adalah jawaban yang **dikumpulkan sebelum batas waktu** yang telah ditentukan.

Jakarta, 20 Februari 2026



Paul Tsai Adinata
Assistant Development Officer

Soal

Case

RUSA

[10 January █████] Climate Change has put the world on a halt! We must move from Earth immediately!

[12 March █████] The Committee of Nations gathered to discuss the future of mankind in space travel.

[27 November █████] The nations have banded together to give the world a second chance. [01 May █████] More

and more nations have joined the RUSA Act, we deserve a fighting chance. [31 July █████]

RUSA has successfully launched the first space station into orbit. [30 November █████]

Man kind's survival? [16 April █████] Retribution for Anomalous events popping

less mission of adventuring the skies. [05 May █████] Weird things are happening less and less, and we might have RUSA to thank for it.

[14 SEPTEMBER █████] I RUSA HAD JUST CELEBRATED ITS 25TH ANNIVERSARY AND THEY ARE STILL UPDATING PAPER. [02 December █████] RUSA has officially achieved their first pragalactic travel! [29 February █████] RUSA has started to digitize itself, they were

seen interview ing their staffs. [23 October █████] RUSA has officially confirmed that they are switching to a more technical oriented approach on their data. [10 December █████] RUSA's Board of Directors have

announced their lead developer on their internal software! [19 December █████] RUSA's lead developer has been tasked with the determination of the requirements of their internal software. they have been provided with a transcript of the interview conducted with the staff at RUSA. are they able to perform up to standard? We'll find out soon enough! [21 December █████] The lead developer is also tasked with the design and programming of the application within RUSA, many have said, the where comes to believe that this one is different, that this one will significantly bring mankind from the dark it once was. (Scribble)

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➤ **Interview Transcript**

- You are to determine **actors, functional requirements, non-functional requirements**, and **use cases** from the provided **interview transcript**.

➤ **Internal Documents**

- The **internal documents** provided will act as a helper to better understand the **motive** and **attributes** of the **use cases** or **requirements**.

➤ **Technological Stack**

- **Tauri – Rust**
- **Svelte – TypeScript**
- **SQL Server / PostgreSQL / MongoDB**
- The use of any **external libraries** must be approved by either the **Assistant Development Officer** or the **TPA Case Maker**.

➤ **Optimization**

- Utilize **Redis Cache** for caching.
- **Rust memory management** must be **optimized**.
- **Avoid N+1 queries**.

➤ **Architectural Pattern**

- **Model View Controller Architecture**

➤ **Design Patterns**

- **State / Event pattern**
- **Singleton pattern**
- Implement **3 more** of any **design patterns** of your choosing.

➤ Security

- Implement **basic authentication functionalities**
 - **Authentication guards (Frontend and Backend).**
 - Only store **Hashed** and **SALTed** passwords.
 - **At least two fields** are provided for **authentication**.
 - You must use **Tauri's state management** to **store** the current **logged in user**.
- Implement **basic authorization functionalities**
 - **Authorization (role) guards (Frontend and Backend)**
- All **database transactions** are to be made from the **Rust/Tauri application layer**.

➤ User Experience

- All the **user experience** and **interface (UX/UI)** must be **clear, representative, and accessible**.
- If an **error occurs**, display the **corresponding** and **relevant error message**.
- The system should employ **fields** based on **specific requirements**; for instance, use a **password field** instead of a **text field** for handling **passwords**.

➤ Code Maintenance

- You must perform basic **clean code practices**.
- You must perform good **GitHub commit practices**.

➤ Data Practices

- All **deletes** must be **soft deletes**.
- **Deletions, Updates, and Insertions** must be **logged**.

➤ Non-Functional Requirements

- **Real-time operations** must not rely on mechanisms such as **database polling**.

➤ General Requirements

- To access the application's core functionality, users are required to go through an **authentication** and **authorization process**.
- To **proxy** something is to **validate** and **possess decision-making authority**.
- **Requests** by default are **voted** upon by **The Directors**, unless stated otherwise.
- **Request for conclusions** are not voted upon; however, they are handled by the **corresponding director(s)** that assigned the **problem**.
- **Budgeting/Funds/Monetary requests or reports** must be **itemized** and **descriptive** and reports must be **accompanied** by an **invoice**, unless stated otherwise.
- As a form of **internal communication between everyone**, utilize an **email-esque** format and layout. **Messages** must be sent and received in **real time** and **notify** users of incoming messages.
 - Format includes **[From]**, **[To]**, **[CC]**, **[BCC]**, **[Subject]**, **Message Recalls**, **Scheduled Messages**, **multiple users** to be sent to at once, and allows for **groups** consisting of a group of users.

➤ Notes

- To ensure a **smooth development process** and a **well-designed system**, you are to design the following **components**,
 - **Use Case Diagram(s)**.
 - Ten (10) Unique **Full Use Case Descriptions**.
 - Ten (10) Unique **Activity Diagrams**.
 - Ten (10) **Multi-layer Sequence Diagrams** with **Fragment(s)**.
 - **Updated Class Diagram** with **Relationships**.
- For better understanding of the system, you are to choose **use cases** with **complex logic** and each use case selection shall be monitored by the **TPA Case Maker**.
- To ensure consistent workflow while designing the diagrams, you are to follow **Satzinger's Design Concepts**.

- You are allowed to make **assumptions** about the **diagrams** and **concepts**, but they must be **logical** and accompanied with **reasoning** and **evidence**, and you must be able to explain all diagrams that you create.
- Your **application** must be developed based on your **diagrams**.
- For the purposes of this **TPA**, the application you develop will only be based on actors determined by the **Case Maker**, make sure to show your **use case diagram** to the Case Maker within the **first week**.
- Make sure to utilize the **RUSA logo** provided.

Good Luck

Confront the challenges of learning and outgrow the boundaries together
Bluejackets 24-2